# BRAINSTORM GROUP (BSG)

## MTH101
## Elementary Mathematics I
## Mock Test Question (2025)

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# Reviewed By

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## SET THEORY

1. In a group of 800 people, 220 are retired soldiers, 600 people have guns, if 60 retired soldiers do not have guns, how many of the rest of people do not have guns?
   A. 140 B. 200
   C. 230 D. 600

2. Which of the following set description is not correct?
   A. $B = set\ geometry\ books$ B. $A = \{1, 2, 3; 4\}$
   C. $D = \{x: x\ is\ even\}$ D. $C = \{a, b, c\}$

3. Which of the following is an empty set?
   A. $\{x: is\ a\ real\ number\ and\ x^2 - 1 = 0\}$
   B. $\{x: is\ a\ real\ number\ and\ x^2 + 1 = 0\}$
   C. $\{x: is\ a\ real\ number\ and\ x^2 - 9 = 0\}$
   D. $\{x: is\ a\ real\ number\ and\ x^2 = x + 2\}$

4. All inhabitants of a certain country speak either French and/or English. If 60% speak French and 50% speak English, what percentage of them speak only one language?
   A. 10% B. 40%
   C. 50% D. 90%

5. In a class of 20 boys, 16 play tennis, 12 play rugby and 2 are not allowed to play games. Find how many students play only rugby.
   A. 2 B. 6
   C. 10 D. 12

6. If $V = \{1, 2, 3\}$ and $W = \blacksquare$, find $V \cap W$
   A. $\{1, 2\}$ B. $\{2, 3\}$
   C. $\{2, 4\}$ D. $\{1, 2, 3\}$

7. If $M \supset N$, then $(M \cap N) \cup N$ is
   A. $\emptyset$ B. M
   C. N D. $\mathcal{U}$

8. If $Q = \{all\ perfect\ squares\ less\ than\ 30\}$ and $P = \{all\ odd\ numbers\ from\ 1\ to\ 10\}$. Find $(Q \cap P)$.
   A. $\{0, 1, 4, 9\}$ B. $\{1, 4, 9\}$
   C. $\{1, 4\}$ D. $\{1, 9\}$

9. List the elements of the set $X = \{n^2 + 1;\ n = 0, 1, 2, 3\}$
   A. $\{0, 2, 4, 9\}$ B. $\{1, 2, 5, 10\}$
   C. $\{0, 1, 2, 3\}$ D. $\{0, 2, 5, 10\}$

10. Find $n(\{1, \{2, 3\}, \{4, 5, 6\}\})$
    A. 1 B. 2
    C. 3 D. 6

11. In an examination written by 30 students, 15 passed MAT101, 12 students passed PHY101, 3 students passed both subject. Find the number of students that passed at least one subject.
    A. 20 B. 21
    C. 24 D. 30

12. Given $n(B - A) = 20, n(A \cup B) = 70\ and\ n(A \cap B) = 25.\ Find\ n(A)$.
    A. 20 B. 25
    C. 45 D. 50

13. The cardinality of the characters in "MISSISSIPPI." is
    A. 4 B. 5
    C. 8 D. 11

14. $[(A \cap A') \cup (A \cup A')]'$ is equivalent to
    A. $\emptyset$ B. A
    C. A' D. $\mathcal{U}$

15. Find the cardinality of $\{n \in \mathbb{N} : -5 < 2n < 12\}$
    A. 5 B. 6
    C. 7 D. 8

16. If $V = \{1, 2, 3\}$ and $W = \{2, 3, 4\}$, find $V \cup W$
    A. $\{2, 4\}$ B. $\{2, 3, 4, 5\}$
    C. $\{1, 2, 3, 4\}$ D. $\{1, 2, 4, 6\}$

17. Which of the following describe a disjoint between set A and B?
    A. $A \cup B = \mathcal{U}$ B. $A \cap B = \mathcal{U}$
    C. $A \cap B = 0$ D. $A \cap B = \emptyset$

18. $[(A \cap \mathcal{U}') \cup (A \cup \emptyset)']'$ is equivalent to
    A. $\emptyset$ B. A C. A' D. $\mathcal{U}$

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19. Let the universal set $\mathcal{U}$ be the set of integers $\mathcal{U} = \{x : 0 < x \leq 10\}$. Find the complement of the set $P = \{x : x \in \mathcal{U}, x \text{ is not divisible by } 4\}$

A. $\{1, 2, 4, 8\}$ B. $\{3,5, 6, 7, 9, 10\}$
C. $\{4, 8\}$ C. $\{1, 2, 3, 5, 6, 7, 9, 10\}$

20. Given $\mathcal{U} = \{a - j\}, A = \{vowels\}$ and $B = \{first\ five\ alphabets\}$, find $n(A \cup B)$

A. $\{a, b, c, d, e, i, o, u\}$ B. $\{a, e, i, o, u\}$
C. $\{a, e, i\}$ D. $\{a, b, c, d\ e, i\}$

21. Out of 26 students, 18 got credits in mathematics, and 17 of them in the group are boys. If 7 girls got credit, how many boys did not get credit?

A. 2 B. 6
C. 8 D. 10

22. Every pupil in a certain class owns either a cap or belt. 46 own cap, 28 own belt and 10 own both. How many pupils are in the class?

A. 54 B. 64
C. 74 D. 84

23. Given $n(A \cap B') = 20, n(A' \cap B) = 30, n(A \cup B) = 60$. If $n(\mathcal{U}) = 75$ find $n(A' \cap B')$

A. 10 B. 15
C. 20 D. 25

24. Given the set $X = \{1,\ 2,\ 3\}$ and set $Y = \{3,\ 1,\ 2\}$, which of the following statement is true for $X$ and $Y$?

A. $X \neq Y$ B. $X \geq Y$
C. $X = Y$ D. $X \equiv Y$

25. Given that U = {x : x is an integer, $2 \leq x \leq 10$}, $A = \{x : x \text{ is a prime number}\}$, $B = \{x: \text{ is a multiple of } 3\}$. List the elements of $A' \cap B'$.

A. {2, 3, 5, 6}
B. {4, 8, 9, 10}
C. {2, 3, 5, 6, 7, 9}
D. {4, 8, 10}

## MAPPING AND FUNCTION

26. The functions $g$ and $f$ are given by $g(x) = 4 - 3x$ and $f(x) = x^2 + ax + b$, where $a$ and $b$ are non-zero constants. Given that $fg(2) = -5$ and $gf(2) = -29$, find the value of $a$ and $b$.

A. $a = -1, b = 4$ B. $a = 1, b = -4$
C. $a = 4, b = 1$ D. $a = 4, b = -1$

27. The function $f$ is given as $f(x) = \frac{x+6}{x+2}, x \in \mathbb{R}, x \neq -2$, solve the equation $f(x) = f^{-1}(x)$

A. $-3, -2$ B. $3, -2$
C. $-3, 2$ D. $3, 2$

28. The function $h$ is given as $h(x) = \frac{x}{x-1}, x \in \mathbb{R}, x \neq 1$, find in its simplest form the composition $hh(x)$

A. $\frac{x-1}{x}$ B. $x - 1$
C. $1 - x$ D. $x$

29. The functions $f$ and $g$ are given by $f(x) = x^2$ and $g(x) = \frac{1}{x+2}$, solve the equation $fg(x) = \frac{4}{9}$

A. $\frac{1}{2}$ and $\frac{7}{2}$ B. $-\frac{1}{2}$ and $\frac{7}{2}$
C. $\frac{1}{2}$ and $-\frac{7}{2}$ D. $-\frac{1}{2}$ and $-\frac{7}{2}$

30. The functions $f$ and $g$ are given by $f(x) = ln(4 - 2x)$ and $g(x) = e^{3x}$, solve the equation $fg(x) = 0$

A. $\frac{1}{2} ln \frac{3}{2}$ B. $\frac{3}{2} ln \frac{1}{2}$
C. $2ln \frac{3}{2}$ D. $-\frac{1}{2} ln \frac{3}{2}$

31. Find the inverse function of $g(a) \to 2e^{2a} - 2$.

A. $ln\left(\frac{a+2}{2}\right)$ B. $2ln\left(\frac{a+2}{2}\right)$
C. $\frac{1}{2} ln(a + 2)$ D. $\frac{1}{2} ln\left(\frac{a+2}{2}\right)$

32. If $f(x) = 2(x - 3)^2 + 3(x - 3) + 4$ and $g(y) = 5 + y$. Find $g[f(3)]$.

A. 4 B. 5
C. 9 D. 0

33. Determine the zeros of the function defined by $h(x) = x^2 - x$

A. 0 and 1 B. 0 and $-1$
C. 1 and $-1$ D. 0 and 2

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34. Given that $f(x) = x^5 + 4x^4 - 6x^2 + 2x + 2$, find $f(-1)$.
A. $-3$ B. $-2$
C. $-1$ D. 1

35. What is the set of values of x for which the mappings $x \to x^2$ and $x \to x^3$ have the same functional values?
A. $\{-1, 0, 1\}$ B. $\{1\}$
C. $\{0, 1\}$ D. $\{2\}$

36. If $f(g(x)) = x^2 + 2x + 1$ and $f(x) = x^2$, find $g(x)$.
A. $x + 2$ B. $x - 2$
C. $x + 1$ D. $x^2 + 2$

37. Determine the zero of the function defined by $h(x) = x^2 + x - 6$.
A. $-2$ B. 3
C. 1 D. 2

38. If every element in the domain has different image in the co-domain, such type of mapping is called
A. constant mapping
B. onto mapping
C. one-one mapping
D. composite mapping

39. A mapping f is called _____ if every element of the co-domain is an image of at least one element in the domain?
A. constant mapping
B. onto mapping
C. one-one mapping
D. composite mapping

40. The rule that assigns an element to two or non-empty set is
A. logic B. set
C. mapping D. rules

41. Given the function $h(x) \to \frac{1}{2}e^{2-x} - 2$, find the inverse function.
A. $2 + ln[2(x-2)]$ B. $ln[2(x+2)] - 2$
C. $2 - ln[2(x-2)]$ D. $2 - ln[2(x+2)]$

42. Two functions $f$ and $g$ are defined on the set $R$ of real numbers by: $f(x) = x^2 - 2x - 1$ and $g(x) = x - 1$. Find the value of x for which $f(x) = g(x) - 2$
A. 0 *or* 1 B. 1 *or* 2
C. 0 *or* 2 D. 1 *or* 3

43. Two functions $f$ and $g$ are defined on the set $R$ of real numbers as follows: $f: x = 2x - 3$ and $g: x = x^2 + 5$. Find $f[g(x-1)]$
A. $2x^2 - 4x + 12$ B. $2x^2 - 4x + 9$
C. $x^2 - 2x + 6$ D. $2x^2 - 2x + 12$

44. Given that image of p under the mapping $x \to 3x + 2$ is $-10$, what is the value of $p$?
A. $-4$ B. $-3$
C. $-2$ D. $-1$

45. The mock test is organized by
A. Brainstorming group (BSG)
B. Brainstorm Group (BSG)
C. Brainstormer Group (BSG)
D. Brainstromy Gorup (BGS)

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| ANSWERS | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. | A | 11. | C | 21. | B | 31. | D | 41. | D |
| 2. | B | 12. | D | 22. | B | 32. | C | 42. | B |
| 3. | B | 13. | B | 23. | B | 33. | A | 43. | B |
| 4. | D | 14. | A | 24. | C | 34. | A | 44. | A |
| 5. | A | 15. | C | 25. | D | 35. | C | 45. | B |
| 6. | B | 16. | B | 26. | D | 36. | C | | |
| 7. | C | 17. | D | 27. | C | 37. | D | | |
| 8. | D | 18. | B | 28. | D | 38. | C | | |
| 9. | B | 19. | C | 29. | D | 39. | B | | |
| 10. | C | 20. | D | 30. | A | 40. | C | | |

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